

Basic Quantitative Method: TA Session Week 1

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TA Session Website

- All materials in the TA Session can be found in [here](#)
 - Including slides, practices, etc.



Today's agenda

- 1 Introduction to TA Session
- 2 Brief Introduction to R
- 3 Download R & RStudio
- 4 Basics of R
 - RStudio Interface
 - Assign Values
 - Data Types
 - Download Packages
 - Help Function
- 5 R Guides & Tips / Reading CSVs
 - Open a script
 - `getwd()`, `setwd()`
 - Project
 - `read.csv()`
 - Folder Organization
 - Coding styles

What will we do in the TA session?

- ① Learn how to use R
 - Basic syntax, Data manipulation, Data visualization, etc.
- ② In-class practice
 - Some practices will be prepared to check your understanding in each TA session.
- ③ R Programming Challenges

How to contact me?

- 1 Sending email
 - Please mail to r13323002@ntu.edu.tw

What is R/RStudio?

- **R** is a free and open-source programming language for **statistical computing** and **data analysis**.
- **RStudio** is an integrated development environment (IDE) for R.



Icon for R



Icon for RStudio

Why choose R?

Aspect	R	Stata	Python
Costs	Free, open-source	Paid	Free, open-source
Primary uses	Data analysis/ Statistics	Data analysis/ Statistics	Various purposes
Learning	Moderate	Easier	Moderate
Typical users	Econ/Stats/Soc. sci.	Econ/Stats/Soc. sci.	Popular in industries

Download R & RStudio

- Download R [here](#).
- Download RStudio [here](#).
- Install R first, and then install RStudio.

Basics of R: RStudio Interface

The screenshot displays the RStudio interface with four main panes:

- Source:** Contains the R script file `ggplot2.R`. The code is as follows:


```
1 library(ggplot2)~
2 mpg_plot <- ggplot(mpg, aes(x = displ, y = hwy)) + ~
3   geom_point(aes(colour = class))~
4 ~
5 mpg_plot|
6
```
- Console:** Shows the execution of the code from the Source pane:


```
> library(ggplot2)
> mpg_plot <- ggplot(mpg, aes(x = displ, y = hwy)) +
+   geom_point(aes(colour = class))
>
> mpg_plot
>
```
- Environments:** Displays the Global Environment with a table of objects:

Name	Type	Len...	Size	Value
mpg_plot	gg	9	29.1...	List of 9
- Plots:** Shows a scatter plot of `hwy` (y-axis) versus `displ` (x-axis). The points are colored by the `class` variable. A legend on the right lists the classes: 2seater, compact, midsize, minivan, pickup, subcompact, and suv.

Basics of R: RStudio Interface

- **Source** (Script): write your code here.
- **Environment**: check your data and objects here.
- **Console**: execute your code here.
- **Files/Plots/Packages/Help**: check your files, plots, packages, and help here.

Basics of R: Assign Values

- `<-` is the **assignment operator** in R. We use it to assign value to a variable.

```
# example  
a <- 3
```

- A value assigned to a variable can be a number, a character, a dataframe, or the result of an expression.

```
# example  
b <- a*a  
# b = 9  
x <- "hello" # x = "hello"
```

Basics of R: Data Types

- There are several basic data types in R:
 - **Numeric**: numbers with decimal points.
 - **Character**: text
 - **Logical**: TRUE or FALSE.
- To check the data type of a variable, use the `class()` function.

```
# example
x <- 5
class(x) # Output: "numeric"
y <- "Hello World!"
class(y) # Output: "character"
z <- TRUE
class(z) # Output: "logical"
```

Basics of R: Data Types

- To store data in R, we have the following ways:
 - **Vector**: one-dimensional sequence of values of the same data type.
 - **List**: one-dimensional sequence of values (could) with different data types.
 - **Matrix**: two-dimensional structure of values of the same data type.
 - **Data frame**: two-dimensional structure of values (could) with different data types across columns.

Basics of R: Data Types (Vector)

- Examples of vector: use `x:y` or `c()` to create vector

```
id <- 1:4 # 1 2 3 4
class(id)
# Output: "integer"
score <- c(30, 86, 50, 95)
class(score) # Output: "numeric"
name <- c("Sam", "John", "Andy", "Judy")
class(name) # Output: "character"
```

Basics of R: Data Types (List)

- Examples of list:

```
my_list <- list("Hello World!", 1, TRUE, score)
```

- Output

```
> print(my_list)
[[1]]
[1] "Hello World!"

[[2]]
[1] 1

[[3]]
[1] TRUE

[[4]]
[1] 30 86 50 95
```

Basics of R: Data Types (Matrix)

- Examples of `matrix(data, nrow=..., ncol=...)`:¹

```
id <- 1:4
score <- c(30, 86, 50, 95)
id_score <- matrix(c(id, score), nrow = 4)
```

- Output

```
> print(id_score)
      [,1] [,2]
[1,]     1  30
[2,]     2  86
[3,]     3  50
[4,]     4  95
```

¹Also try what happens when without `nrow` and when `nrow = 2`!

Basics of R: Data Types (Data Frame)

- Examples of `data.frame()`:

```
df <- data.frame(stu_id = id,  
                  stu_score = score)
```

- Output

```
> print(df)  
  stu_id stu_score  
1      1        30  
2      2        86  
3      3        50  
4      4        95
```

Basics of R: Download Packages

- To download and install a package, use the `install.packages()` function.

```
install.packages("tidyverse")
```

- To load a package into your R session, use the `library()` function.

```
library(tidyverse)
```

- Add `"` when use `install.packages()` install, but no need when use `library()`.

Basics of R: Help Function

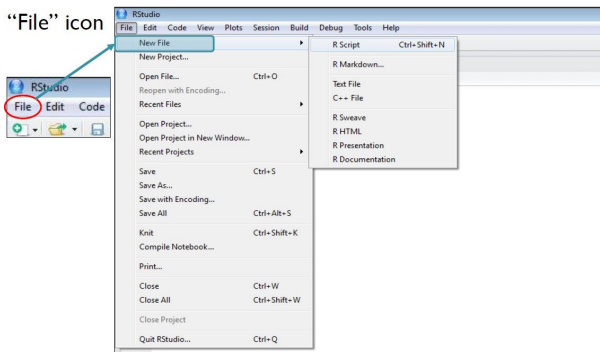
- To get help on a specific function, use the `help()` function or the `?` operator to check the documentation for the function.

```
help(mean)  
?mean
```

- This will open a help page with information about the function, its arguments, and examples of usage.
- Other types of help function: `help.start()` and `help.search()`.

Script in RStudio

- A **script** is a text file that you can write your code in.



Why should we open a script?

- Main Reason: We open an R script so we can save it!
- Save your code in the script such that it is executable and reproducible.

The screenshot shows the RStudio interface. The main editor window displays an R script with the following code:

```

1
2 # clear the memory, import package for graphing
3
4 rm(list = ls())
5 library(ggplot2)
6
7
8 # 1. import the dataset
9 loan <- read.csv("data/loans_full_schema.csv")
10 county <- read.csv("data/county.csv")
11
12 # 2. Create contingency table object: 'table()'
13
14 ## using 'table' function to easily build contingency table within dataset
15 table <- table(loan$application_type,
16               loan$homeownership)
17
18 print(table)
19 class(table)
20
21 ## we can also specify the column name in the function arguments
22 ## by giving a vector of names to parameter 'dnn', which means "dimension names"
23 table <- table(loan$application_type,
24               loan$homeownership,
25               dnn = c("application type",
26                       "home ownership"))
26
27
28 
```

The Environment pane on the right shows the objects in the global environment: Data (bar_graph, bar_graph_withlabs, county, loan, scatter_plot, scatter_plot_scale, table_wtthsum, table), Values (table), and Files (Zoom, Exp). The Plots pane shows a scatter plot of 'homeownership' (y-axis) versus 'application type' (x-axis). The console at the bottom shows the output of the script:

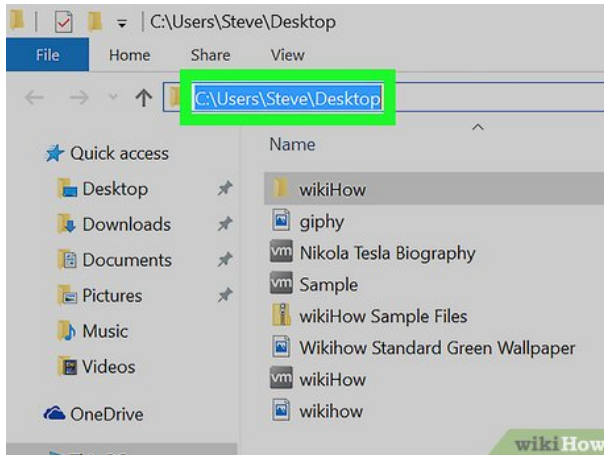
```

R 4.0.4 - C:/Users/Manny/OneDrive/Graduate/統計學/TA/2025/R code/
> ggplot(loan) + aes(application_type, homeownership)
> print(scatter_plot)
> scatter_plot_scale <- scatter_plot +

```

Path of a file

- A **path** for a file is the place where you save it.



Get/Set Current Directory

- `getwd()`: Get current working directory

```
getwd() #Example Output: "C:/your/working/directory"
```

- `setwd()`: Change your working directory²

```
setwd("C:\\your\\working\\directory") #Error!  
setwd("C:\\\\your\\\\working\\\\directory") #Works on Windows!  
setwd("C:/your/working/directory") #Works everywhere!
```

²You can use `dir.exists()` to check if the directory exists.

RStudio project

- **RStudio project:** A tool to organize all files related to a task.
 - Sets the root folder as the wd for all R sessions within the project.
 - No more `setwd()`!
 - Keep history, options ... etc.
 - More advantages. e.g. portable path: `here()` function.
- Open a project by (File → New Project) at the top left corner of RStudio.
- When should I open a RStudio project?
 - If the task lasts for multiple periods, or with multiple files.
- Recommend to open a RStudio project for the TA Session Practices!

Read CSV File: read.csv()

• Syntax:

```
df <- read.csv("file.csv") #Work if file in working directory
df <- read.csv("folder_name/file.csv") #If file is in folder
df <- read.csv("C:/your/working/directory/file.csv")
```

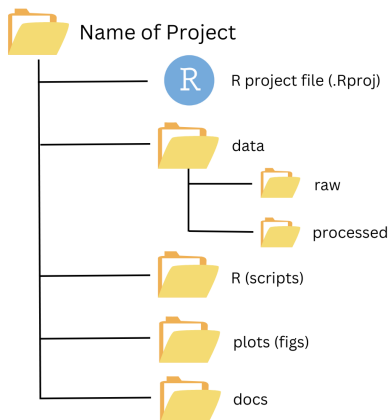
• Example:

```
babies <- read.csv("C:/your/working/directory/babies.csv")

# Do this if you have your R project created!
earthquake <- read.csv("data/earthquake.csv")
```

Folder Organization

- After opening a project, it is important to have well-organized folders.
- Different types of things (scripts, data, output, etc.) should be saved in different folders!



Coding styles: Naming

- Only lowercase letters, numbers, and `_` within a name.
- The name should be readable!³

```
# Good
lesson_one
lesson_1

# Bad
LessonOne
lessonone
one
a <- 1
```

³See [this website](#) for more on coding styles.

Coding styles: Spacing

- Always put a space after a comma, never before.
- Do not put spaces inside or outside parentheses.

Good

```
x[, 1]
```

```
mean(x, na.rm = TRUE)
```

Bad

```
x[,1]
```

```
x[ ,1]
```

```
x[ , 1]
```

```
mean (x, na.rm = TRUE)
```

```
mean( x, na.rm = TRUE )
```

Coding styles: Function Indents

- Indent the argument name with a single indent (two spaces).
- Don't put + or , in a new line!

Good

```
long_function_name <- function(  
  a = "a long argument",  
  b = "another argument",  
  c = "another long argument"  
) {  
}
```

Bad

```
long_function_name <- function(a = "a long argument",  
  b = "another argument",  
  c = "another long argument"  
) {  
}
```

```
long_function_name <- function(a = "a long argument", b = "another argument",  
  c = "another long argument") {  
}
```

Practice Session

- We will have some R practices after each lecture.
 - Practices can be found in the TA Session's Website: [here](#).
- Raise your hand/come to TA if you finish or have any questions on the practices.
 - Feel free to leave after finishing them.